COLLEGE OF ENGINEERING, GUINDY
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Railway Digital Twin with Wildlife Safety

An Educational Simulation Integrating Godot 4, MQTT,
a Trusted Node.js Control Plane and ESP32 Edge Sensing

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Abstract

Train–animal collisions remain a persistent operational risk on forested and rural railway corridors while conventional level-crossing protection offers the same single mechanism for every kind of intruder. This project develops an educational *Railway Digital Twin with Wildlife Safety*: a complete, runnable miniature of a modern wildlife-protected railway line in which the same event can be observed simultaneously from four cooperating surfaces — a three-dimensional twin simulation, a web operations dashboard, a trusted backend control plane, and an ESP32 edge controller, each communicating over an authenticated IoT message bus.

The system is built as a staged software stack. **Godot 4.7** renders a two-track "stadium" railway world (UP and DOWN loops of 1253.98 m and 1228.85 m around a 250 m straight-sided footprint) with an island station, interlocked crossovers, a wildlife level crossing with boom barriers, carved river with road and railway bridges, catenary, dynamic signal masts and Libre-TrainSim-derived Blender train, station, deer and elephant assets. **Node.js 22+** with Express and SQLite is the trusted control plane: the sole publisher of railway commands, state snapshots and alerts, enforcing bcrypt/JWT authentication, role-based access control (viewer/operator/admin), command-envelope validation, anti-replay protection through a persistent SQLite registry, and a full audit trail. **Eclipse Mosquitto 2** brokers the machine traffic on MQTT TCP 1883/WebSocket 9001 with per-client credentials and least-privilege topic ACLs. A **React 18/Vite** dashboard presents live state, per-block signal aspects, per-train speed control, meet scheduling, wrong-line routing and hazard evidence badges. An **ESP32 (Arduino/PlatformIO)** node carries fail-safe field inputs — wildlife and sensor-fault pulls, protective outputs, debounce, SNTP-gated clock truth and non-blocking telemetry. A **FastAPI ML microservice** using torchvision's SSDLite MobileNetV3 detector confirms animal sightings from twin-camera frames behind a two-frame evidence policy.

The wildlife protection chain implements a staged escalation NONE → SUSPECTED (30 km/h limit) → CONFIRMED (0 km/h, STOP/RED) driven by trackside sensor huts, onboard simulated cameras and ESP32 edge reports, with backend-authoritative hazard fusion, recovery-latched resets and restrictive fail-safe behaviour on any loss of

communication. Every subsystem is verified by an automated gate — 42/42 Godot suites, 141/141 backend tests, 8/8 ML tests, 16/16 ESP32 native tests and a clean web build — plus live end-to-end scenarios including a chance-based meet scheduling, an interlocked wrong-line run and the two-frame camera confirmation chain end-to-end on the live stack. The system is emphatically a classroom simulation and not certified railway signalling equipment.

Keywords: Digital Twin, Railway Safety, Wildlife, MQTT, IoT, ESP32, Godot Engine, Node.js, React, Machine Learning

Contents

List of Figures

List of Tables